

Phase 1 Report

NLP and Digital Preservation Annotation Platform

Team Members:

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Target Languages:

Assamese (Primary) , Bodo , Karbi , Dimasa etc.

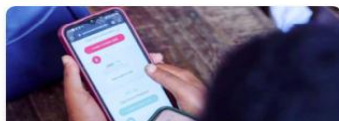
1.Comparative Study of Existing Platforms :

1.1 [Mozilla Common Voice](#) :

Common Voice is the gold standard for open-source, community-led data creation. It separates contributions into Scripted Speech (reading sentences aloud), Spontaneous Speech (responding to prompts, useful for oral-first languages), and Language Text (submitting public-domain text), and is fully open for anyone to download.

Common Voice is a free, open source platform for community-led data creation

Anyone can preserve, revitalise and elevate their language by sharing, creating and curating text and speech datasets.



Scripted Speech

Read sentences aloud in your language and contribute to the most diverse public participation speech dataset in the world.

Speak >



Spontaneous Speech

Respond to prompts to create datasets for organic, colloquial contexts. Perfect for oral-first languages.

Answer >



Language Text

Create or share public domain prompts, sentences, and text for translation, small language models, and more.

Add Text >

*Common Voice's three contribution modes: **Scripted Speech**, **Spontaneous Speech**, and **Language Text**.*

Common Voice

moz://a

[Speak](#) [Listen](#) [Write](#)

Click  then read the sentence aloud

He was chairman of the North Carolina House
Finance Committee.



Common Voice

moz://a

[Speak](#) [Listen](#) [Write](#)

Add [a public domain sentence](#)

Sentences contributed here will be added to a publicly available

Sentence

i like cultivating

Sentence Domain

Select up to three domains (optional)

Agriculture and Fo... ×

Citation

Reference the source of your sentence (required)

[Speak](#) [Listen](#) [Write](#)

By area, Quebec is the largest province of
Canada.



YES



NO

Features of common voice:

Speak: This mode is for contributing "Scripted Speech," where users read specific sentences aloud. It is the traditional method of data collection for the project, where a prompt appears on the screen and the user records themselves reading it. The project also includes "Spontaneous Speech," where users respond to open-ended prompts to create more natural, colloquial datasets.

Write: This feature allows users to contribute text for the dataset. Users can create or share public domain prompts and sentences that will later be used by others for the "Speak" phase. This also involves reviewing transcriptions to ensure the text corpus remains high-quality.

Listen: This feature involves validating audio clips recorded by other community members. Users listen to the recording and verify that it matches the corresponding text. This quality-assurance process ensures the dataset is accurate enough to train AI algorithms.

Common Voice Scripted Speech 26.0 - Assamese

 License: CC0-1.0

 Steward: Common Voice

 Dataset ID: cmqi6ss2y00guo507u3jsgp4i 

 Task: ASR  Release Date: 6/17/2026  Format: MP3  Size: 160.40 MB

Download



Files

Latest Version

 common-voice-scripted-... 160.40 MB

Point of Contact

Mozilla Foundation
commonvoice@mozilla.com

Description

A collection of read speech recordings in Assamese (অসমীয়া).

Specifics

Licensing

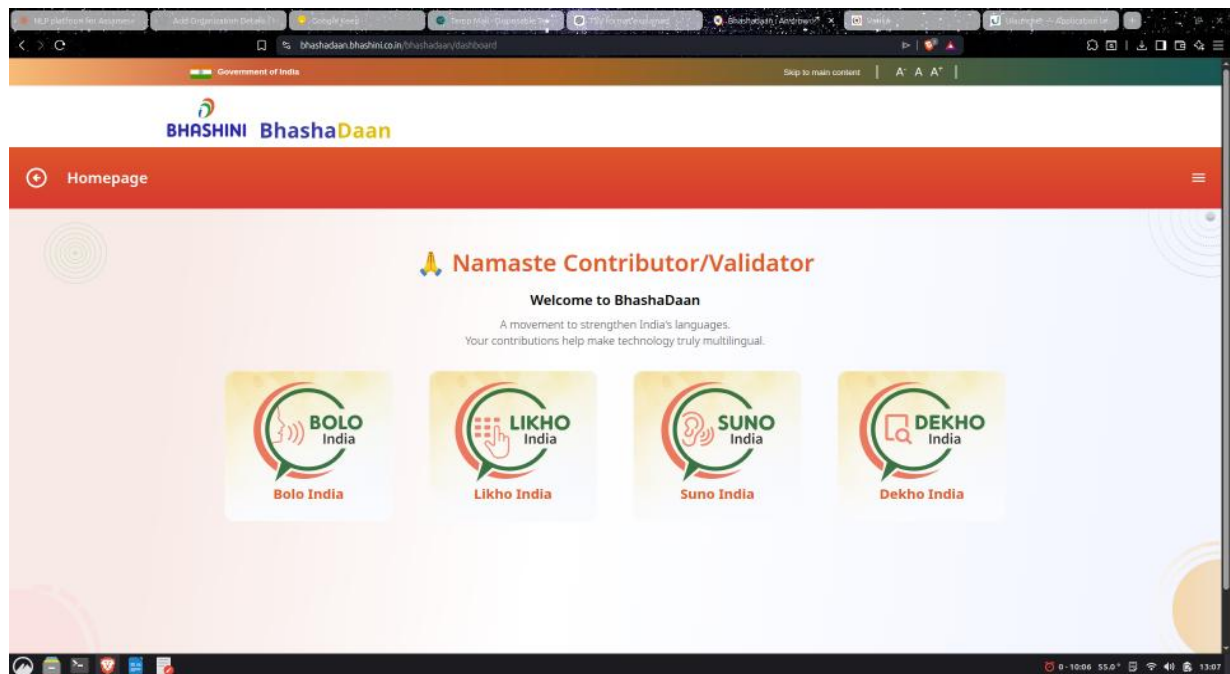
Creative Commons Zero v1.0 Universal (CC0-1.0)

<https://spdx.org/licenses/CC0-1.0.html>

Assamese dataset on common voice

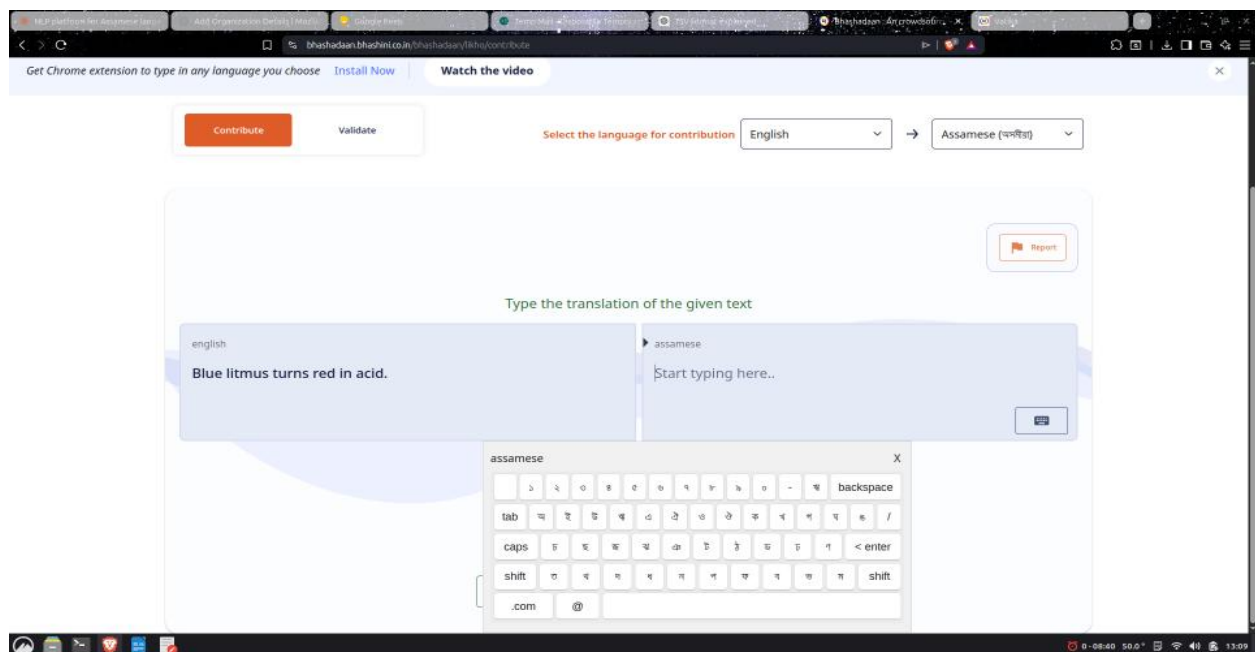
1.2 Bhashini — BhashaDaan (Government of India)

Bhashini is India's National Language Translation Mission. Its crowdsourcing arm, BhashaDaan, organizes contribution into four task types — Bolo (speak), Likho (write/translate), Suno (listen), and Dekho (see) — giving volunteers a simple, gamified entry point for donating language data across India's regional languages.



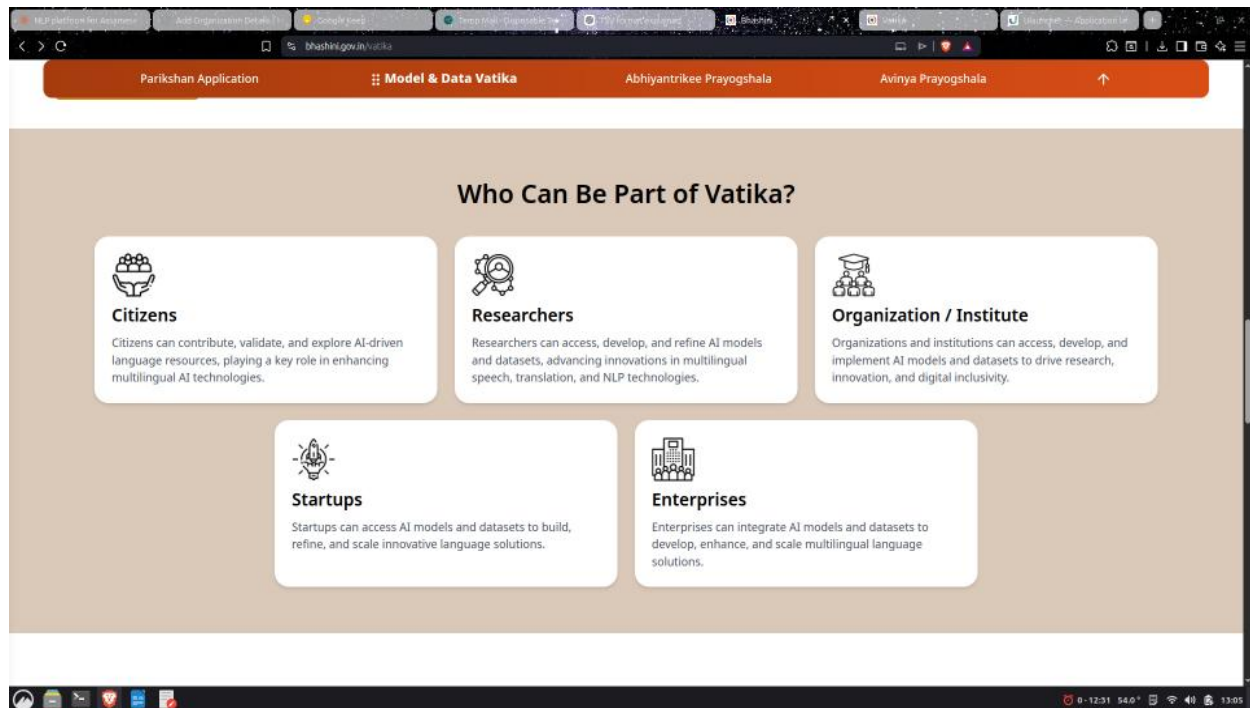
BhashaDaan homepage: contribution is split into Bolo, Likho, Suno, and Dekho India tasks.

The Likho (write) module shows a direct translation-contribution interface, complete with a native Assamese virtual keyboard so contributors without an Assamese-enabled device can still type accurately.



Likho India contribution screen: translating an English sentence into Assamese using the built-in keyboard.

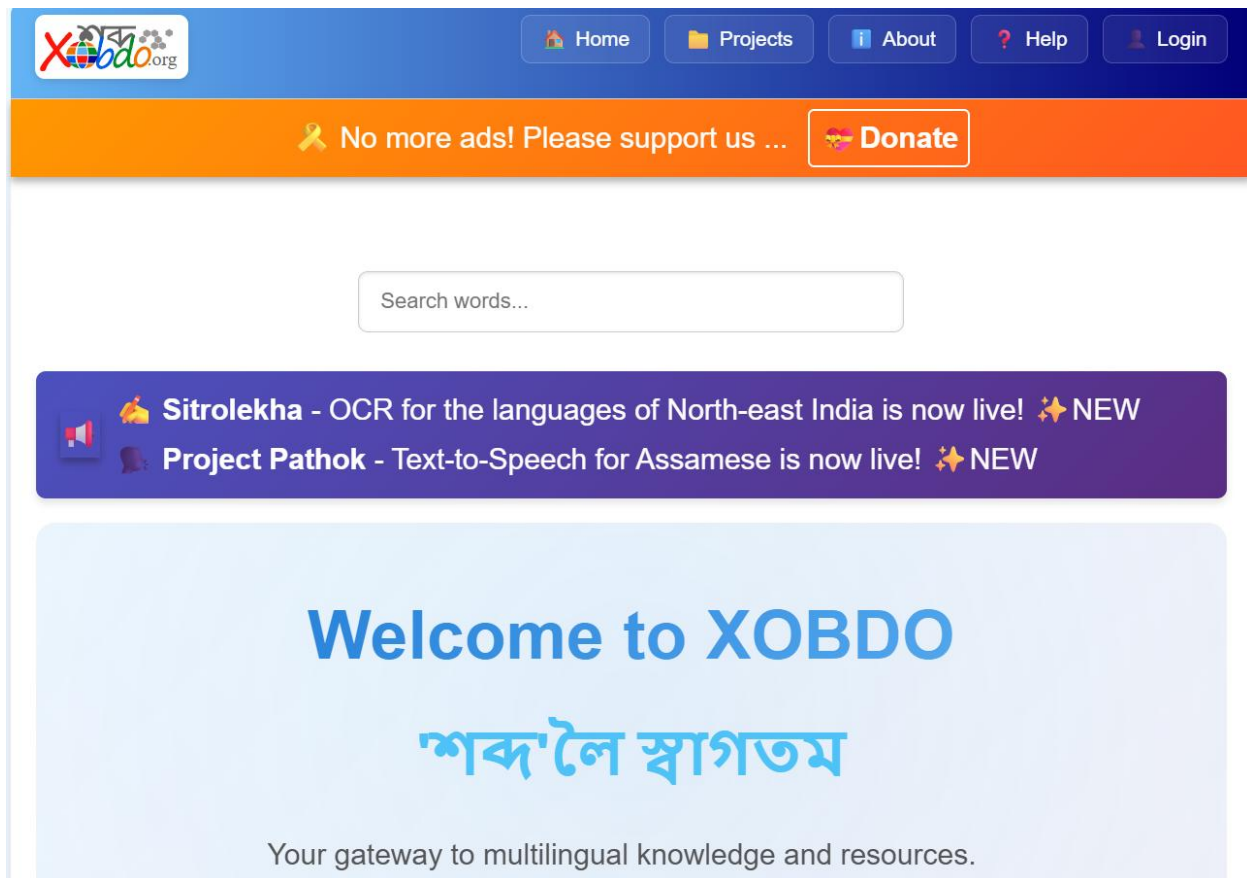
Bhashini also runs **Vatika**, a broader model-and-dataset marketplace open to citizens, researchers, organizations, startups, and enterprises — illustrating an institutional model for distributing the verified datasets a platform like ours could eventually export.



Vatika's stakeholder categories: Citizens, Researchers, Organizations, Startups, and Enterprises.

1.3 Xobdo (Assamese Community Dictionary & Language Platform)

XOBDO is a community-driven language platform developed primarily for Assamese and other North-East Indian languages. Unlike Mozilla Common Voice, which focuses on collecting speech datasets, or Bhashini, which collects multilingual AI datasets, XOBDO focuses on building digital language resources through community participation. It combines dictionary development with AI-based language tools, making it one of the most relevant reference platforms for Assamese language preservation.



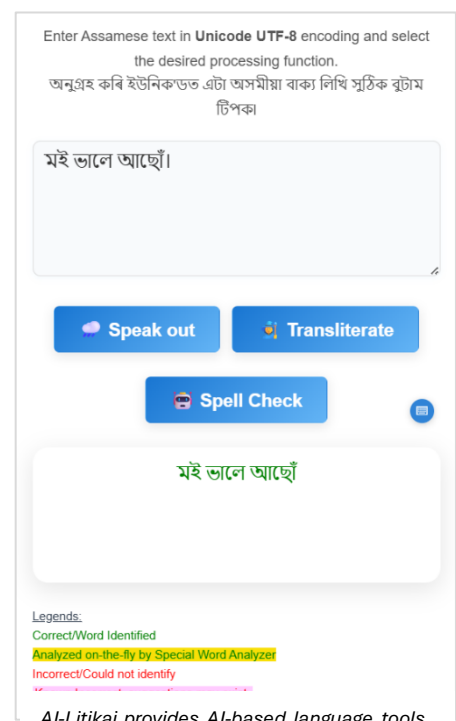
XOBDO homepage showcasing its community-driven dictionary and language resource platform.

AI Language Tools

Apart from dictionary development, XOBDO provides several AI-powered language tools for Assamese through its AI-Litikai initiative. These tools include:

- Spell Checker
- Text-to-Speech (TTS)
- Speech-to-Text (SrutoLipi)
- Transliteration
- Optical Character Recognition (OCR)

These applications demonstrate how community-contributed language resources can be utilized for building practical NLP systems for low-resource languages.



AI-Litikai provides AI-based language tools

Speech-to-Text (SrutoLipi)

XOBDO is also developing **SrutoLipi**, a Speech-to-Text system for Assamese. It converts spoken Assamese into text and is designed to support different dialects and real-world speech. This illustrates how crowdsourced language resources can contribute to the development of speech recognition technologies for regional languages.



SrutoLipi – Assamese Speech-to-Text interface developed under the XOBDO project.

 **Coming Soon**

Currently under development

SrutoLipi (শ্রুতলিপি) is an innovative Assamese dictation and speech-to-text tool being developed using state-of-the-art artificial intelligence technologies. This advanced system will enable accurate voice recognition and transcription specifically optimized for the Assamese language.

Editorial Verification

To maintain the quality and reliability of its dictionary, XOBDO follows an **editorial verification process**. Community members can submit new words, meanings, or corrections, but these contributions are **reviewed by editors before being published**. This ensures that the information added to the platform is accurate, consistent, and suitable for long-term language preservation.

Key Observations

The study of XOBDO provides several useful insights for our proposed annotation platform:

- *Community participation is an effective approach for building language resources.*
- *Editorial verification improves the reliability of contributed content.*
- *AI tools can be developed once sufficient high-quality language data is available.*
- *Regional language preservation requires continuous contribution from native speakers.*
- *A single platform can integrate dictionary resources with speech and NLP technologies.*

Although XOBDO successfully supports Assamese language preservation through dictionaries and AI tools, it primarily focuses on dictionary development. Our proposed platform extends this idea by collecting structured datasets including phrase translations, audio recordings, dialect metadata, syntax annotations, and a verification workflow, making the collected data more suitable for future AI and NLP research.

1.4 Summary

Comparative Summary

Platform	Primary Focus	Verification Model	Key Takeaway
BhashaDaan (Bhashini)	Indian regional languages, govt-backed	Structured review via institution	Task-type segmentation (Bolo/Likho/Suno/Dekho) for gamified UX
Mozilla Common Voice	Open-source multilingual speech	Peer validation, open dataset release	Three contribution modes incl. spontaneous speech for oral languages
Xobdo	Community-driven dictionary & language resources	Editor review before publishing	Demonstrates how community participation can preserve regional languages while supporting AI applications.

2. Proposed Database Schema

The schema is designed around four core relational tables, plus a JSONB payload for fine-grained syntax annotation. It is built to support phrases, translations, dialect metadata, and a structured verification workflow that the platforms above largely lack.

2.1 Users

Column	Type	Description
user_id	PK, Integer	Unique identifier for each contributor
username / email	Text	Display name and login credential
native_dialect	Text	e.g. Kamrupi, Goalpariya, Standard Assamese
trust_score	Integer	Gamification metric based on approved submissions

2.2 Raw_Phrases

Column	Type	Description
phrase_id	PK, Integer	Unique identifier for the source sentence
text_content	Text	The original Assamese phrase
source_type	Text	Literature, News, Spoken, User-submitted
dialect_metadata	Text	Regional variant of the source phrase

2.3 Annotations

Column	Type	Description
annotation_id	PK, Integer	Unique identifier for the submitted work
phrase_id / user_id	FK	Links to Raw_Phrases and Users
task_type	Text	Translation, Syntax Tagging, Audio Recording

content	JSONB	Translation text or structured POS/dependency tags
status	Text	Pending, Approved, Rejected

2.4 Verifications

Column	Type	Description
verification_id	PK, Integer	Unique identifier for the review action
annotation_id / user_id	FK	Links the reviewer to the annotation reviewed
vote	Integer	+1 correct, -1 incorrect
timestamp	Datetime	When the verification was cast

2.5 Handling Complex Syntax (JSONB Payload)

Rather than splitting every token into separate rows, the content column in Annotations stores a compact JSON object capturing the root, part-of-speech tag, morphology, and dependency relation — preserving Assamese's agglutinative structure for direct export to ML pipelines.

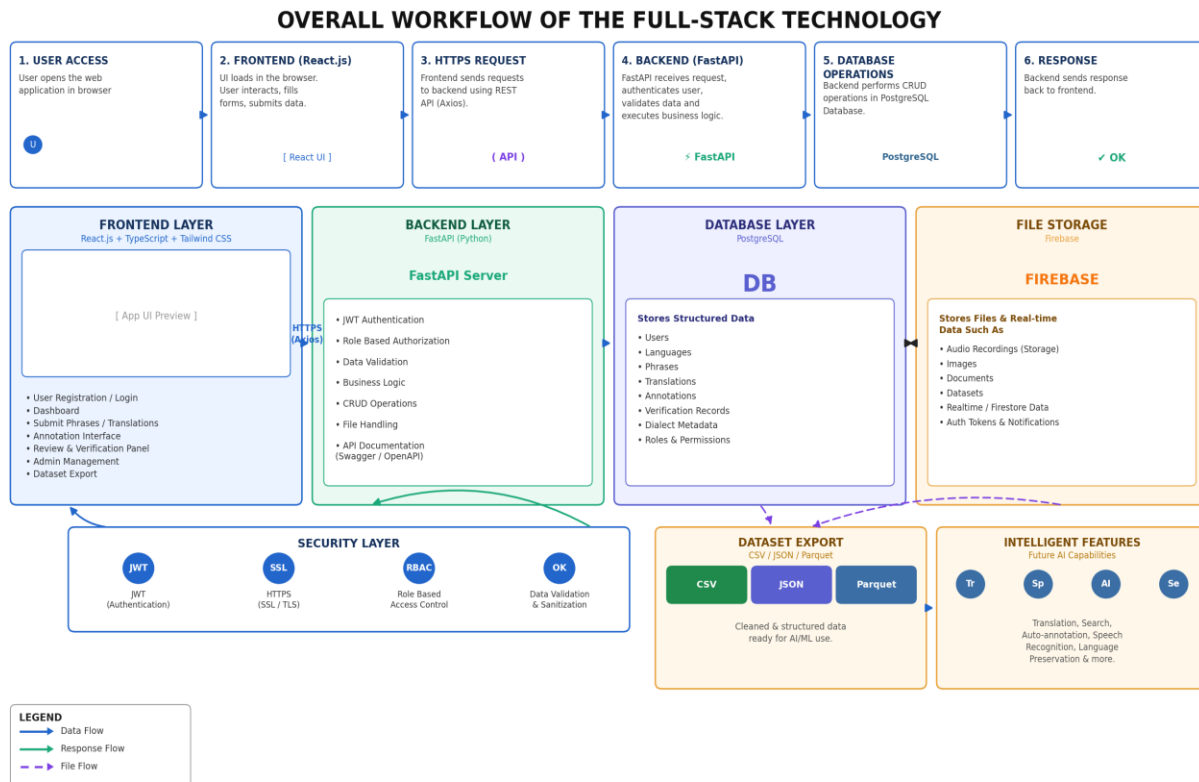
```
[{"token_id":1,"original_word":"মানুহজনক","root":"মানুহ","pos_tag":"NOUN","morphology":{"classifier":"জন","case":"ACCUSATIVE"},"dependency":{"head_token_id":3,"relation":"OBJECT"}}]
```

This design allows an instant, clean export: e.g. “all Assamese-to-English translations with status = Approved and at least 3 positive verifications.”

3. Recommended Full-Stack Architecture

The platform is built on four core technologies: React.js for the contributor-facing portal, FastAPI for business logic and APIs, PostgreSQL for structured relational data, and Firebase for files and real-time

data. The diagram below summarizes the full request lifecycle and supporting layers.



Overall workflow: request lifecycle, the four core layers, the security layer, and downstream dataset export.

3.1 End-to-End Request Flow

A single user action moves through six stages: (1) User Access — the contributor opens the web app; (2) Frontend (React.js) — the UI loads and the user fills a form or submits data; (3) HTTPS Request — the frontend calls the backend over a REST API via Axios; (4) Backend (FastAPI) — the request is authenticated, validated, and processed; (5) Database Operations — FastAPI performs CRUD operations against PostgreSQL; (6) Response — the result is returned to the frontend and rendered for the user.

3.2 Frontend Layer — React.js + TypeScript + Tailwind CSS

The client-facing application that contributors and validators interact with directly.

- User registration and login
- Contributor dashboard
- Submit phrases and translations
- Annotation interface (syntax tagging, audio recording)
- Review and verification panel
- Admin management console
- Dataset export controls

3.3 Backend Layer — FastAPI (Python)

The API and business-logic layer that mediates every interaction between the frontend and the data stores.

- JWT authentication
- Role-based authorization (RBAC)
- Data validation
- Core business logic for annotation and verification workflows
- CRUD operations against the database
- File handling for audio/image uploads
- Auto-generated API documentation (Swagger / OpenAPI)

3.4 Database & File Storage

Data Storage Split

Layer	Technology	Stores
Structured Data	PostgreSQL	Users, languages, phrases, translations, annotations, verification records, dialect metadata, roles and permissions
Files & Real-time Data	Firebase	Audio recordings, images, documents, datasets, realtime/Firestore data, auth tokens and notifications

3.5 Security Layer

These four protections apply across every request rather than at a single stage, and wrap both the frontend–backend and backend–database boundaries.

- **JWT:** JWT — authentication
- **SSL:** SSL / TLS — encrypted transport over HTTPS
- **RBAC:** RBAC — role-based access control
- **Validation:** Validation — data validation and sanitization

3.6 Dataset Export & Intelligent Features

Once data is verified, it can be exported in standard formats and used to power AI features further down the pipeline.

- **Dataset Export:** Export formats: CSV, JSON, and Parquet — cleaned, structured data ready for AI/ML use
- **Intelligent Features:** Future AI capabilities: translation, search, auto-annotation, speech recognition, and broader language preservation tooling built on top of the exported dataset

End of Phase 1 Report.